Overview of Multivariate Methods

Principal Component Analysis

Partial Least Squares Regression

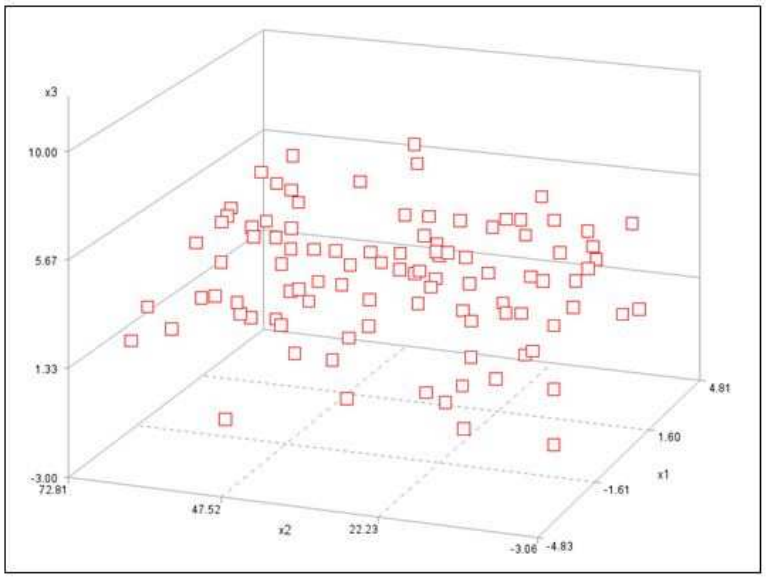
Cluster Analysis

Discriminant Analysis

Canonical Correlation Analysis

Data Analysis in High Dimensions

Prof. Dr. Christina Andersson



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What Is Multivariate Statistics?

- Multivariate data analysis refers to any statistical technique used to analyze data that arises from more than one variable.

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What Is Multivariate Statistics?

- Multivariate analysis is the area of statistics that deals with observations made on many variables.
- The main objective is to study how the variables are related to one another, and how they work in combination.
- The analysis of multivariate data is of concern in every research discipline: biology, medicine, environmental science, sociology, economics, education, linguistics, archaeology, anthropology, psychology and behavioural science.

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What Is Multivariate Statistics?

- Univariate statistics:
  One variable is studied.
- Bivariate statistics:
  Two variables are studied simultaneously
- Multivariate statistics:
  More than two variables are studied simultaneously

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Where Is Multivariate Statistics Used?

- All natural and physical processes are essentially multivariate in nature.
- The challenge is to understand the process in a multivariate way, where variables are connected and their relationships understood, as opposed to a bunch of univariate processes, i.e. single variables at a time, isolated from one another.

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Where Is Multivariate Statistics Used?

- Consumer and market research
- Quality control and quality assurance across a range of industries such as food and beverage, paint, pharmaceuticals, chemicals, energy, telecommunications, etc
- Process optimization and process control
- Research and development

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What Can You Do with Multivariate Statistics?

- Identify the dominant patterns in the data, such as groups, outliers, trends, and so on.
- Analyze groups in the table, how these groups differ, and to which group individual table rows belong.
- Use one set of variables (columns) to predict another, for the purpose of optimization, and to find out which columns are important in the relationship.
- Identify homogeneous subgroups of cases or variables based on some measure of "distance".

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Examples of Multivariate Methods

- Principal components analysis
- Partial least squares regression
- Cluster analysis
- Discriminant function analysis
- Canonical variate analysis

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Pros with Multivariate Methods

- Richer realistic design
- Accounting for dependencies

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Cons with Multivariate Methods

- Larger sample size are often required
- More difficult to interpret
- Less known about the robustness of assumptions

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Eigenvector and Eigenvalue

An *eigenvector* of a square matrix A is a non-zero vector $\vec{x}$ such that

$$A\vec{x} = \lambda\vec{x}$$

for some number λ called an *eigenvalue* of A. The vector $\vec{x}$ is called an eigenvector corresponding to the eigenvalue λ.

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Finding the Eigenvalues of A

To find the eigenvalues of A, we find the roots of

$$det(A - \lambda I) = 0$$

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Finding the Eigenvectors of A

To find the eigenvectors corresponding to eigenvalue λ, we solve the linar system

$$(A - \lambda I)\vec{x} = 0$$

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Example

The matrix

$$A = \begin{pmatrix} 1 & 4 \\ 9 & 1 \end{pmatrix}$$

has the characteristic equation:

$$\left| \begin{pmatrix} 1 & 4 \\ 9 & 1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \right| = 0$$

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Example Continued

This means that

$$\begin{vmatrix} 1-\lambda & 4 \\ 9 & 1-\lambda \end{vmatrix} = 0$$

We get the eigenvalues $\lambda_1 = -5$ and $\lambda_2 = 7$.

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- Objectives in multivariate statistics
- Matrix algebra revisited

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Background

- With a large number of variables, the data matrix may be too large to study and interpret properly.
- There would be too many pairwise correlations between the variables to consider.
- Graphical display of data may also not be of particular help incase the data set is very large.
- With 12 variables, there will be more than 200 three-dimensional scatterplots to be studied!

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Objective

- To interpret the data in a more meaningful form, it is therefore necessary to reduce the number of variables to a few, interpretable linear combinations of the data.
- Each linear combination will correspond to a principal component.

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Basics

- Principal components are constructed as mathematical transformations of the input variables.
- The first principal component is constructed in such a way that it captures as much of the variation in the input data as possible.
- The second principal component is orthogonal to the first principal component.
- The second principal component captures as much as possible of the variation in the input data not captured by the first principal component.
- And so on ...

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Starting Point

Let

$$\vec{X} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{pmatrix}$$

be a random vector with variance-covariance matrix

$$var(\vec{X}) = \Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1p} \\ \sigma_{21} & \sigma_{22}^2 & \cdots & \sigma_{2p} \\ \vdots & \vdots & \vdots & \vdots \\ \sigma_{p1} & \sigma_{p2} & \cdots & \sigma_p^2 \end{pmatrix}$$

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Starting Point

Consider the linear combinations

$$\begin{aligned} PC_1 &= a_{11}X_1 + a_{12}X_2 + \cdots + a_{1p}X_p \\ PC_2 &= a_{21}X_1 + a_{22}X_2 + \cdots + a_{2p}X_p \\ &\vdots \\ PC_p &= a_{p1}X_1 + a_{p2}X_2 + \cdots + a_{pp}X_p \end{aligned}$$

Each of these components can be seen as a linear regression, predicting PC_i from $X_1, X_2, \ldots, X_p$.

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Variance of PC_i

Note that PC_i is a function of random data, i.e. PC_i is also random and has the variance

$$var(PC_i) = \sum_{k=1}^{p}\sum_{l=1}^{p} a_{ik}a_{il}\sigma_{kl} = \vec{a_i'}\Sigma\vec{a_i}$$

where we summarize a_{ij} into the vector

$$\vec{a_i} = \begin{pmatrix} a_{i1} \\ a_{i2} \\ \dots \\ a_{ip} \end{pmatrix}$$

Covariance of PC_i

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Moreover PC_i and PC_j will have the covariance

$$cov(PC_i, PC_j) = \sum_{k=1}^{p} \sum_{l=1}^{p} a_{ik} a_{jl} \sigma_{kl} = \vec{a_i'} \Sigma \vec{a_j}$$

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Intuitive: First Principal Component PC_1

- The first principal component is the linear combination of x-variables that has maximum variance (among all linear combinations).
- This means that it accounts for as much variation in the data as possible.
- Specifically we will define coefficients $a_{11}, a_{12}, \ldots, a_{1p}$ for that component in such a way that its variance is maximized, subject to the constraint that the sum of the squared coefficients is equal to one.
- This constraint is required so that a unique answer may be obtained.

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Formally: First Principal Component PC_1

More formally, select $a_{11}, a_{12}, \ldots, a_{1p}$ that maximizes

$$var(PC_1) = \sum_{k=1}^{p} \sum_{l=1}^{p} a_{1k} a_{1l} \sigma_{kl} = \vec{a_1'} \Sigma \vec{a_1}$$

subject to the constraint that

$$\vec{a_1'} \vec{a_1} = \sum_{j=1}^{p} a_{1j}^2 = 1$$

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Second Principal Component PC_2

The second principal component is the linear combination of x-variables that accounts for as much of the remaining variation as possible, with the constraint that the correlation between the first and second component is zero.

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Second Principal Component PC_2

Select $a_{21}, a_{22}, \ldots, a_{2p}$ that maximizes

$$var(PC_2) = \sum_{k=1}^{p}\sum_{l=1}^{p} a_{2k}a_{2l}\sigma_{kl} = \vec{a_2'}\Sigma\vec{a_2}$$

subject to the constraint that

$$\vec{a_2'}\vec{a_2} = \sum_{j=1}^{p} a_{2j}^2 = 1$$

along with the additional constraint that these two components will be uncorrelated with one another:

$$cov(PC_1, PC_2) = \sum_{k=1}^{p}\sum_{l=1}^{p} a_{1k}a_{2l}\sigma_{kl} = \vec{a_1'}\Sigma\vec{a_2} = 0$$

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i^{th} Principal Component PC_i

All subsequent principal components have the same properties:

- They are linear combinations that account for as much of the remaining variation as possible.
- They are not correlated with the other principal components.

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i^{th} Principal Component PC_i

Select $a_{i1}, a_{i2}, \ldots, a_{ip}$ that maximizes

$$var(PC_i) = \sum_{k=1}^{p}\sum_{l=1}^{p} a_{ik}a_{il}\sigma_{kl} = \vec{a_i'}\Sigma\vec{a_i}$$

subject to the constraint that

$$\vec{a_i'}\vec{a_i} = \sum_{j=1}^{p} a_{ij}^2 = 1$$

along with the additional constraint that the different components will be uncorrelated with one another:

$$cov(PC_i, PC_j) = \sum_{k=1}^{p}\sum_{l=1}^{p} a_{ik}a_{jl}\sigma_{kl} = \vec{a_i'}\Sigma\vec{a_j} = 0$$

i.e. all principal components are uncorrelated with one another.

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How Can We Find the Coefficients?

How do we find the coefficients a_{ij} for a principal component?

The solution involves the eigenvalues and eigenvectors of the variance-covariance matrix Σ.

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How Can We Find the Coefficients?

- We are going to let λ_1 through λ_p denote the eigenvalues of the variance-covariance matrix Σ.
- These are ordered so that λ_1 has the largest eigenvalue and λ_p is the smallest:

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$$

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How Can We Find the Coefficients?

We are also going to let the vectors $\vec{a_1}$ through $\vec{a_p}$

$$\vec{a_1}, \vec{a_2}, \ldots, \vec{a_p}$$

denote the corresponding eigenvectors.

It turns out that the elements for these eigenvectors will be the coefficients of our principal components.

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How Can We Find the Coefficients?

The variance for the ith principal component is equal to the ith eigenvalue:

$$var(PC_i) = var(a_{i1}X_1 + a_{i2}X_2 + \ldots + a_{ip}X_p)$$

How Can We Find the Coefficients?

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Moreover, the principal components are uncorrelated with one another:

$$cov(PC_i, PC_j) = 0$$

The Spectral Decomposition Theorem

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The spectral decomposition theorem tells that the variance-covariance matrix may be written as a function of the eigenvalues and their corresponding eigenvectors.

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The Spectral Decomposition Theorem

The variance-covarianc matrix can be written as the sum over the p eigenvalues, multiplied by the product of the corresponding eigenvector times its transpose as shown in the first expression below:

$$\Sigma = \sum_{i=1}^{p} \lambda_i \vec{a}_i \vec{a}_i' \approx \sum_{i=1}^{k} \lambda_i \vec{a}_i \vec{a}_i'$$

The approximation is useful if $\lambda_{k+1}, \lambda_{k+2}, \ldots, \lambda_p$ are small.

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The Total Variation of $\vec{X}$

- The total variation of $\vec{X}$ is defined as the trace of the variance-covariance matrix, i.e. the sum of the variances of the individual variables.
- This is also equal to the sum of the eigenvalue:

$$\text{trace}(\Sigma) = \sigma_1^2 + \sigma_2^2 + \cdots + \sigma_p^2 = \lambda_1 + \lambda_2 + \cdots + \lambda_p$$

This will give us an interpretation of the components in terms of the amount of the full variation explained by each component.

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Proportion of Variation Explained by the ith Principal Component

- The proportion of variation explained by the ith principal component is then going to be defined to be the eigenvalue for that component divided by the sum of the eigenvalues.
- In other words, the ith principal component explains the following proportion of the total variation:

$$\frac{\lambda_i}{\lambda_1 + \lambda_2 + \cdots + \lambda_p}$$

Proportion of Variation Explained by the First k Principal Components

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A related quantity is the proportion of variation explained by the first k principal components:

$$\frac{\lambda_1 + \lambda_2 + \cdots + \lambda_k}{\lambda_1 + \lambda_2 + \cdots + \lambda_p}$$

Dimension Reduction

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If the proportion of variation explained by the first k principal components is large, then not much information is lost by considering only the first k principal components.

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Sample Variance-Covariance Matrix

- All of this was defined in terms of the population variance-covariance matrix Σ, which is unknown.
- We can estimate Σ by the sample variance-covariance matrix which is given with the standard formula:

$$S = \frac{1}{n-1} \sum_{i=1}^{n} (\vec{X}_i - \vec{\bar{x}})(\vec{X}_i - \vec{\bar{x}})'$$

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Procedure

1. Compute the eigenvalues $\hat{\lambda}_1, \hat{\lambda}_2, \ldots, \hat{\lambda}_p$ of the sample variance-covariance matrix S, and the corresponding eigenvectors $\vec{\hat{a}}_1, \vec{\hat{a}}_2, \ldots, \vec{\hat{a}}_p$.
2. Define the estimated principal components as follows:

$$\begin{aligned}
\hat{PC}_1 &= \hat{a}_{11}X_1 + \hat{a}_{12}X_2 + \cdots + \hat{a}_{1p}X_p \\
\hat{PC}_2 &= \hat{a}_{21}X_1 + \hat{a}_{22}X_2 + \cdots + \hat{a}_{2p}X_p \\
&\vdots \\
\hat{PC}_p &= \hat{a}_{p1}X_1 + \hat{a}_{p2}X_2 + \cdots + \hat{a}_{pp}X_p
\end{aligned}$$

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Dimension Reduction

Generally, we only retain the first k principal component. Here we must balance two conflicting desires:

1. To obtain the simplest possible interpretation, we want k to be as small as possible.
   - If we can explain most of the variation just by two principal components then this would give us a much simpler description of the data.
   - The smaller k is the smaller amount of variation is explained by the first k components.
2. To avoid loss of information, we want the proportion of variation explained by the first k principal components to be large.

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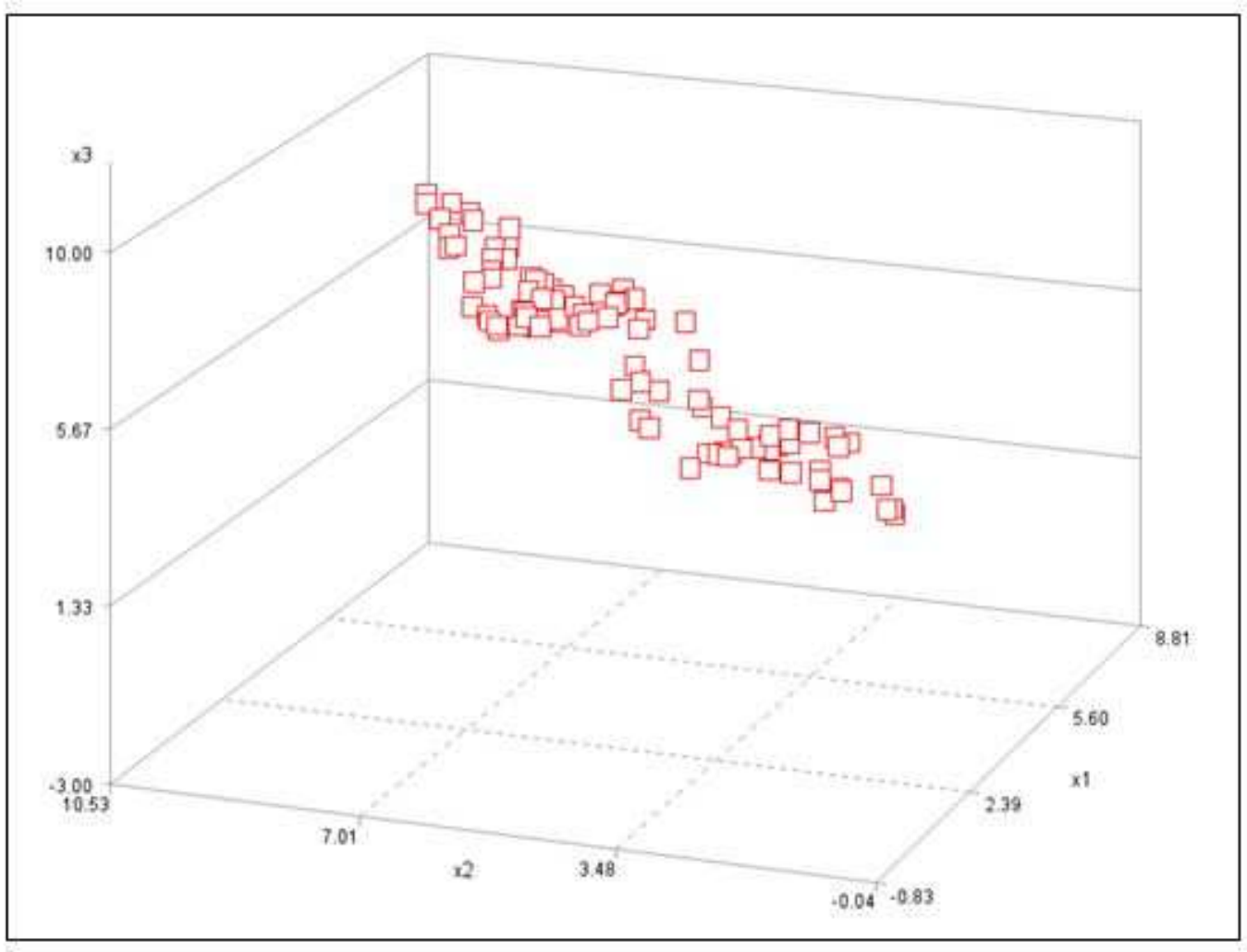
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Example 0:a



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Example 0:a

Eigenvalues of the Correlation Matrix				
	Eigenvalue	Difference	Proportion	Cumulative
1	2.21358154	1.44403082	0.7379	0.7379
2	0.76955072	0.75268297	0.2565	0.9944
3	0.01686775		0.0056	1.0000

Eigenvectors			
	Prin1	Prin2	Prin3
x1	0.650940	0.263685	0.711862
x2	0.645235	0.301851	-.701825
x3	-.399937	0.916164	0.026348

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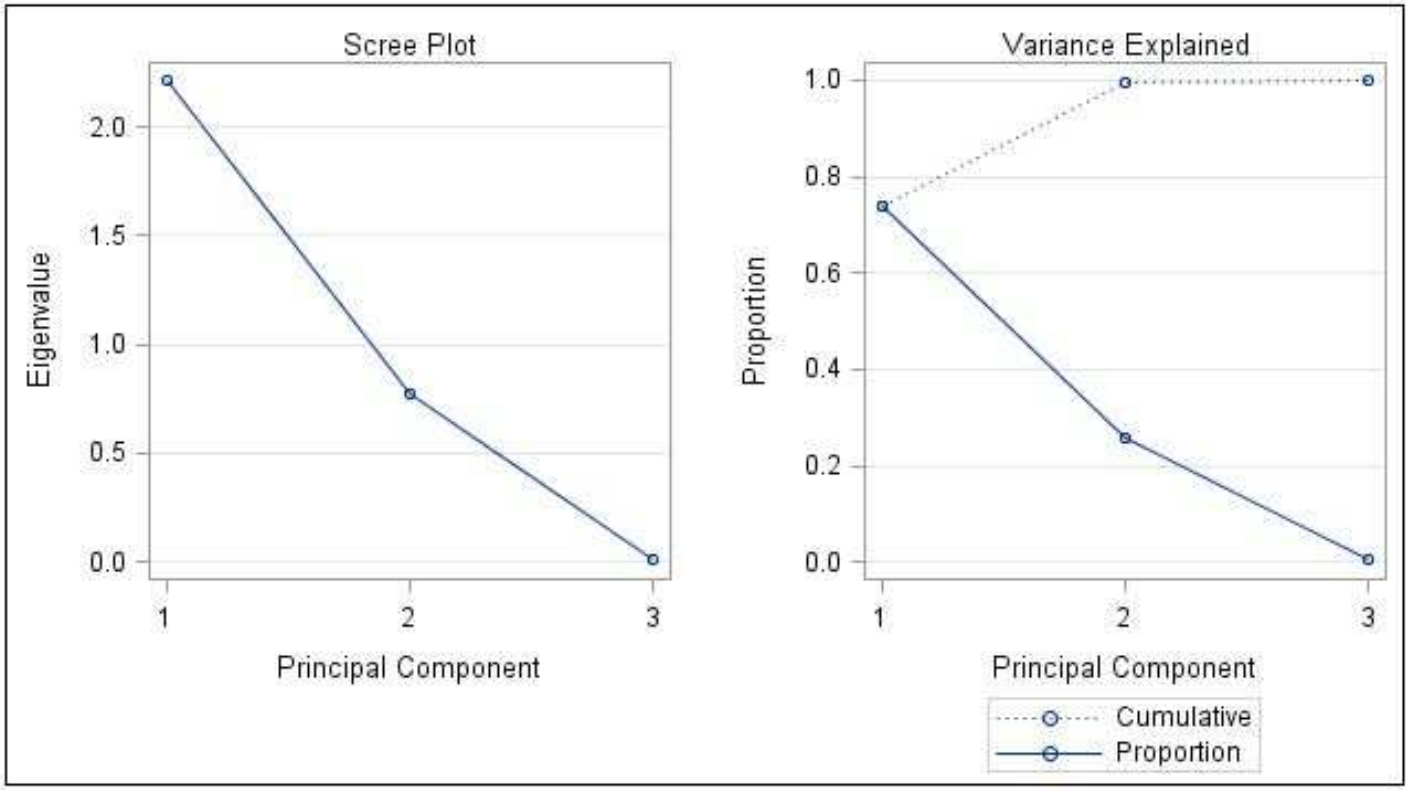
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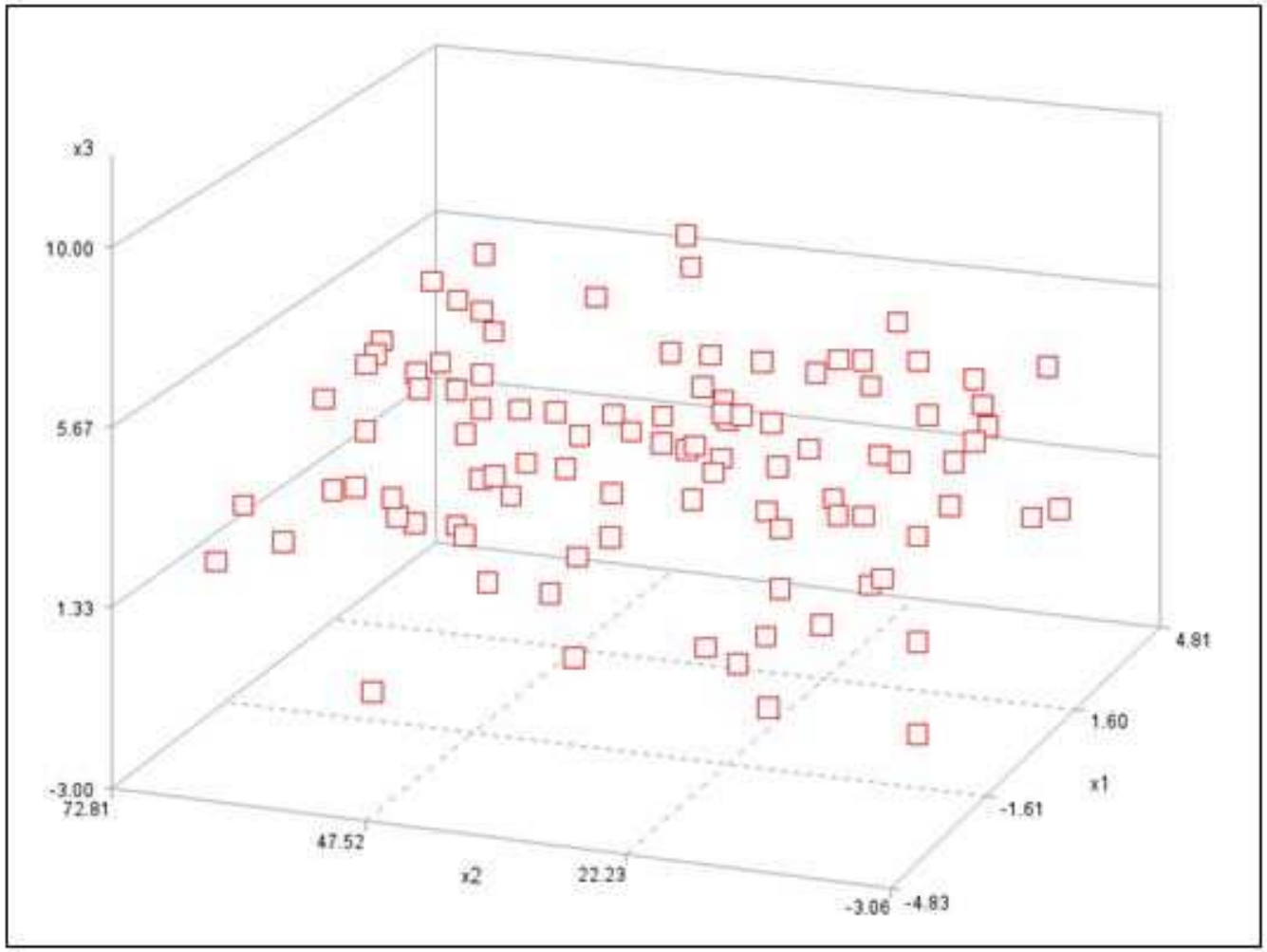
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Example 0:b



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Example 0:b

Eigenvalues of the Correlation Matrix				
	Eigenvalue	Difference	Proportion	Cumulative
1	1.24205697	0.23274599	0.4140	0.4140
2	1.00931098	0.26067894	0.3364	0.7505
3	0.74863205		0.2495	1.0000

Eigenvectors			
	Prin1	Prin2	Prin3
x1	0.704619	-.156546	0.692102
x2	0.708942	0.113759	-.696032
x3	0.030228	0.981097	0.191139

Example 0:b

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Principal Component Analysis

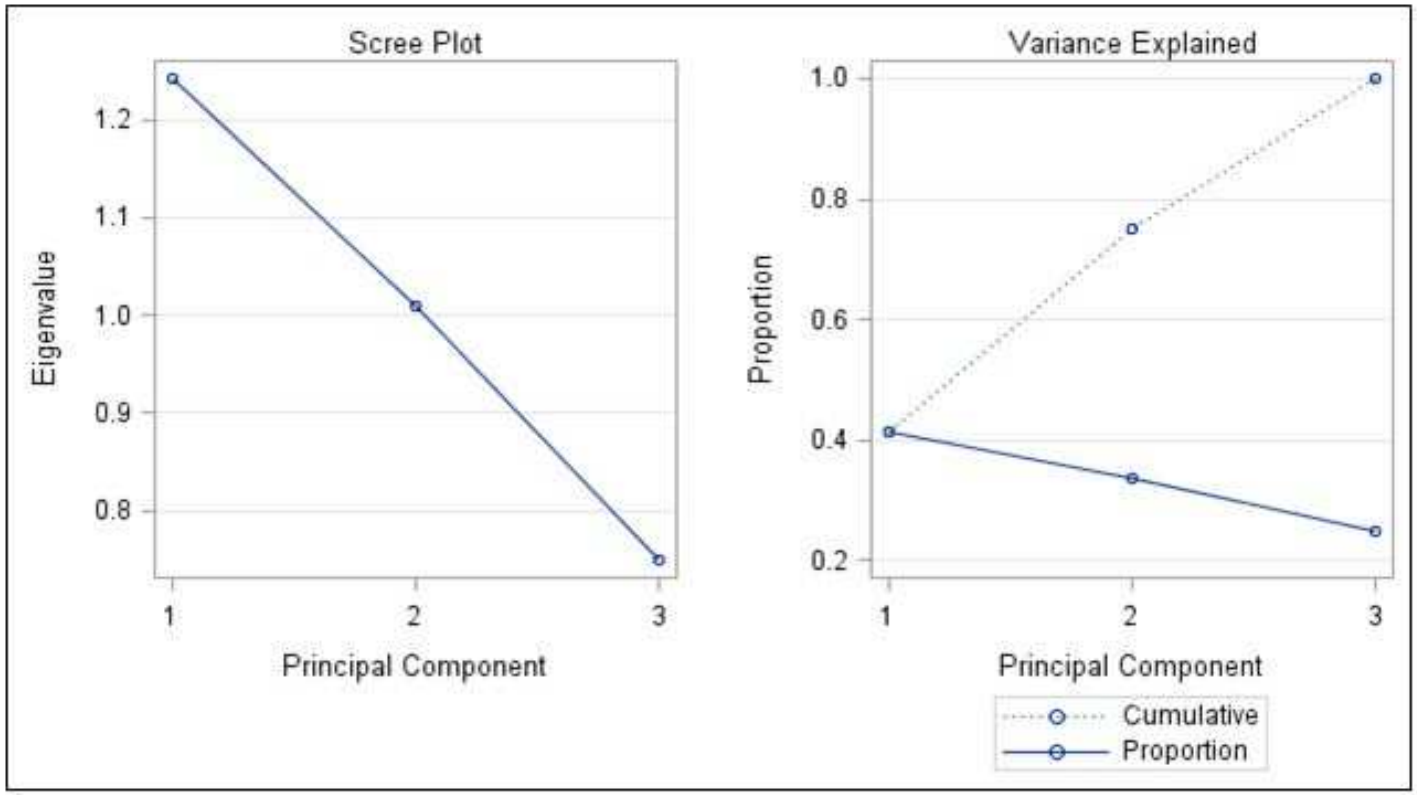
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Example: Places Rated

In the Places Rated data set (Almanac, Boyer and Savageau) 329 communities were rated according to the following nine criteria:

- Climate and terrain
- Housing
- Health care & the environment
- Crime
- Transportation
- Education
- The arts
- Recreation
- Economics

Note that within the dataset, except for housing and crime, the higher the score the better. For housing and crime, the lower the score the better.

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Example: Places Rated

Strategy:

1. Initial data exploration
2. Determine how many principal components should be considered
3. Compute the principal component scores
4. Interpret each principal component

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PCA: Standardize the Variables

Until now, we looked at principal components analysis applied to the raw data. In our earlier discussion we noted that if the raw data is used principal component analysis will tend to give more emphasis to those variables that have higher variances than to those variables that have very low variances.

- The results of principal component analysis depend on the scales at which the variables are measured.
- Variables with the highest sample variances will tend to be emphasized in the first few principal components.
- Principal component analysis using the covariance function should only be considered if all of the variables have the same units of measurement.

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PCA: Standardize the Variables

- If the variables either have different units of measurement (i.e., pounds, feet, gallons, etc), or if we wish each variable to receive equal weight in the analysis, then the variables should be standardized before a principal components analysis is carried out.
- Standardize the variables by subtracting its mean from that variable and dividing it by its standard deviation:

$$Z_{ij} = \frac{X_{ij} - \bar{x}_j}{s_j}$$

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PCA: Standardize the Variables

- The variance-covariance matrix of the standardized data is equal to the correlation matrix for the unstandardized data.
- Therefore, principal component analysis using the standardized data is equivalent to principal component analysis using the correlation matrix.

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PCA: Standardize the Variables

1. The principal components are first calculated by obtaining the eigenvalues $\hat{\lambda}_1, \hat{\lambda}_2, \ldots, \hat{\lambda}_p$ of the correlation matrix R, and the corresponding eigenvectors $\vec{\hat{a}}_1, \vec{\hat{a}}_2, \ldots, \vec{\hat{a}}_p$.
2. Then the estimated principal components scores are calculated using formulas similar to before, but instead of using the raw data we will use the standardized data in the formulae below:

$$\begin{aligned}
\hat{PC}_1 &= \hat{a}_{11}Z_1 + \hat{a}_{12}Z_2 + \cdots + \hat{a}_{1p}Z_p \\
\hat{PC}_2 &= \hat{a}_{21}Z_1 + \hat{a}_{22}Z_2 + \cdots + \hat{a}_{2p}Z_p \\
&\vdots \\
\hat{PC}_p &= \hat{a}_{p1}Z_1 + \hat{a}_{p2}Z_2 + \cdots + \hat{a}_{pp}Z_p
\end{aligned}$$

Rest of the procedure and the interpretations are as discussed before.

Example: Places Rated with Standardized Variables

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Strategy as in the previous example, but based on standardized varables.

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Summary

In this unit we learned about ...

- The objectives of a principal components analysis;
- The construction of the principal components;
- How to interpret the principal components;
- How to select the number of principal components to be considered;
- How to choose between doing the analysis based on the variance-covariance matrix or the correlation matrix.

Partial Least Squares Regression

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Background

- Partial least squares (PLS) is another method for constructing predictive models when the factors are many and highly collinear.
- Note that the emphasis is on predicting the responses and not necessarily on trying to understand the underlying relationship between the variables.

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Typical Applications

- PLS was developed in the 1960's by Herman Wold as an econometric technique, but some of its most avid proponents are chemical engineers and chemometricians.
- PLS has also been applied to monitoring and controlling industrial processes, since a large process can easily have hundreds of controllable variables and dozens of outputs.

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Basics

- In a regression analysis, there may be a huge amount of input variables, but still only a few underlying or latent factors that account for most of the variation in the response.
- The general idea of PLS is to try to extract these latent factors, accounting for as much of the manifest factor variation as possible while modeling the response variable well.
- For this reason, the acronym PLS has also been taken to mean "projection to latent structure."

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Comparison with PCA

PCA

- The factors are chosen to explain as much of the input space variation as possible.
- This approach yields informative directions in the input space, but not in the response space.

PLS

- The factors are chosen simlar to in PCA, but in addition so that the relationship between input space and response space is taken into account.

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Comparison with PCA

PCA

- Spectral decomposition of $X'X$.

PLS

- Spectral decomposition of $X'Y$.

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Example: Olive Oil

A data set with 5 quality measurements (chemical variables, X-variables) and 6 panel sensory panel variables (sensory variables, Y-space) made on 16 olive oil samples.

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Summary

In this unit we learned about ...

- The objectives of a partial least squares regression;
- The similarities and differences of PLS to PCA;
- How to perform a PLS analysis.

Cluster Analysis

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What Is Cluster Analysis?

Cluster analysis is a data exploration technique for dividing a multivariate dataset into "natural" clusters (groups).

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What Is Cluster Analysis?

- We use the cluster analysis to explore whether previously undefined clusters (groups) may exist in the dataset.

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Example 1: Marketing

- A marketing department may wish to use survey results to sort its customers into categories.
- For example, those likely to be most receptive to buying a product, those most likely to be against buying a product, and so forth.

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The Use of Cluster Analysis

- Cluster analysis is used when we believe that the sample units come from an unknown number of distinct populations or sub-populations.
- We also assume that the sample units come from a number of distinct populations, but there is no apriori definition of those populations.
- Our objective is to describe those populations using the observed data.

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Example 2: Woodyard Hammock Data

- Woodyard Hammock, a beech-magnolia forest in northern Florida.
- The data involve counts of the numbers of trees of each species in $n = 72$ sites.
- A total of 31 species were identified and counted, however, only $p = 13$ most common species were retained.

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Example 2: Woodyard Hammock Data

Code	Scientific name	Common name
carcar	Carpinus caroliniana	Ironwood
corflo	Cornus florida	Dogwood
faggra	Fagus grandifolia	Beech
ileopa	Ilex opaca	Holly
liqsty	Liquidambar styraciflua	Sweetgum
maggra	Magnolia grandiflora	Magnolia
nyssyl	Nyssa sylvatica	Blackgum
ostvir	Ostrya virginiana	Blue Beech
oxyarb	Oxydendrum arboreum	Sourwood
pingla	Pinus glabra	Spruce Pine
quenig	Quercus nigra	Water Oak
quemic	Quercus michauxii	Swamp Chestnut Oak
symtin	Symplocus tinctoria	Horse Sugar

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Objective with The Cluster Analysis

- We are to group sample sites together into clusters that share similar species compositions as determined by some *measure of association*.

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Measures of Association

There are several options to measure association. Two common measures:

1. Measure of association between sample units: We need some way to measure how similar two subjects or objects are to one another.
2. Measure of association between clusters: How similar are two clusters?

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Cluster Approaches

Two main approaches of cluster analysis:

- Hierarchical methods
- Non-hierarchical methods

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Hierarchical Methods

- Agglomerative hierarchical algorithms:
  - These methods start by defining each data point to be a cluster.
  - Then, the two closest clusters (to each other) are combined into a new cluster.
  - In each subsequent step, two existing clusters are merged into a single cluster.
- Divisive hierarchical algorithms:
  - These methods start by putting all data points into a single cluster.
  - Then we divide this cluster into two clusters.
  - At each subsequent step, we divide an existing cluster into two clusters.

Comments concerning Hierarchical Methods

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- Agglomerative methods are used much more often than divisive methods.
- Hierarchical methods can be adapted to cluster variables rather than observations. This is a common use for hierarchical methods.

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Non-Hierarchical Methods

- In a non-hierarchical method, the data are initially partitioned into a set of K clusters.
- This may be a random partition or it may be a partition based on a first "good" guess at seed points which form the initial centers of the clusters.
- Then data points are iteratively moved into different clusters until there is no sensible reassignment possible.
- The initial number of clusters (K) might either be specified by the user or by the software algorithm.
- The most commonly used non-hierarchical method is MacQueen's K-means method.

Standard Notation

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- X_{ik} = Response for variable k in sample unit i (the number of individual species k at site i)
- n = Number of sample units
- p = Number of variables

Required Properties for Measures of Association

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- Symmetry
- Positivity
- Identity
- Triangle inequality

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Symmetry

$$d(\vec{X}_i, \vec{X}_j) = d(\vec{X}_j, \vec{X}_i)$$

i.e., the distance between subject one and subject two must be the same as the distance between subject two and subject one.

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Positivity

$$d(\vec{X}_i, \vec{X}_j) > 0 \text{ if } \vec{X}_i \neq \vec{X}_j$$

i.e., the distances must be positive - negative distances are not allowed!

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Identity

$$d(\vec{X}_i, \vec{X}_j) = 0 \text{ if } \vec{X}_i = \vec{X}_j$$

i.e., the distance between the subject and itself should be zero.

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Triangle Inequality

$$d(\vec{X}_i, \vec{X}_k) \leq d(\vec{X}_i, \vec{X}_j) + d(\vec{X}_j, \vec{X}_k)$$

This follows from geometric consideration, where we learnt that sum of two sides of a traingle cannot be smaller than the third side.

Some Common Measures of Association

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- Euclidean distance
- Minkowski distance
- Canberra metric
- ...

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Euclidean Distance

$$d(\vec{X}_i, \vec{X}_j) = \sqrt{\sum_{k=1}^{p}(X_{ik} - X_{jk})^2}$$

- This is used most commonly.
- This is the square-root of the sum of the squared differences between the measurements for each variable.

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Minkowski Distance

$$d(\vec{X}_i, \vec{X}_j) = \left[\sum_{k=1}^{p} |X_{ik} - X_{jk}|^m \right]^{1/m}$$

- Here the square is replaced with raising the difference by a power of m and instead of taking the square root, we take the mth root.

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Canberra Metric

$$d(\vec{X}_i, \vec{X}_j) = \sum_{k=1}^{p} \frac{|X_{ik} - X_{jk}|}{X_{ik} - X_{jk}}$$

Similarity

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For each of these distance measures, the smaller the distance, the more similar (more strongly associated) are the two subjects.

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Notation

- Observations from cluster 1: $X_1, X_2, \ldots, X_k$
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- Distance between a subject with observation vector $\vec{x}$ and a subject with observation vector $\vec{y}$: $d(\vec{x}, \vec{y})$

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Some Common Linkage Methods

- Single linkage
- Complete linkage
- Average linkage
- Centroid method
- Ward's method

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Single Linkage

- In single linkage, we define the distance between two clusters to be the minimum distance between any single data point in the first cluster and any single data point in the second cluster.
- On the basis of this definition of distance between clusters, at each stage of the process we combine the two clusters that have the smallest single linkage distance.

$$d_{12} = \min_{i,j} d(\vec{X}_i, \vec{Y}_i)$$

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Complete Linkage

- In complete linkage, we define the distance between two clusters to be the maximum distance between any single data point in the first cluster and any single data point in the second cluster.
- On the basis of this definition of distance between clusters, at each stage of the process we combine the two clusters that have the smallest complete linkage distance.

$$d_{12} = \max_{i,j} d(\vec{X}_i, \vec{Y}_i)$$

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Average Linkage

- In average linkage, we define the distance between two clusters to be the average distance between data points in the first cluster and data points in the second cluster.
- On the basis of this definition of distance between clusters, at each stage of the process we combine the two clusters that have the smallest average linkage distance.
- This method involves looking at the distances between all pairs and averages all of these distances.
- This is also called UPGMA - Unweighted Pair Group Mean Averaging.

$$d_{12} = \frac{1}{kl} \sum_{i=1}^{k} \sum_{j=1}^{l} d(\vec{X}_i, \vec{Y}_i)$$

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Centroid Method

- In centroid method, the distance between two clusters is the distance between the two mean vectors of the clusters.
- At each stage of the process we combine the two clusters that have the smallest centroid distance.

$$d_{12} = d(\vec{\bar{x}}_i, \vec{\bar{y}}_i)$$

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Ward's Method

- This method does not directly define a measure of distance between two points or clusters.
- It is an ANOVA based approach. At each stage, those two clusters merge, which provides the smallest increase in the combined error sum of squares from one-way univariate ANOVAs that can be done for each variable with groups defined by the clusters at that stage of the process.

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Example: Woodyard Hammock Data

1. Comparison of different cluster approaches
2. Cluster descriptions

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Introduction to Ward's Method

- This is an alternative approach for performing cluster analysis.
- Basically, it looks at cluster analysis as an analysis of variance problem, instead of using distance metrics or measures of association.
- But it is still a hierarchical method.
- It will start out at the leaves and work its way to the trunk, so to speak. It looks for groups of leaves that form into branches, the branches into limbs and eventually into the trunk.
- Ward's method starts out with n clusters of size 1 and continues until all the observations are included into one cluster.
- This method is most appropriate for quantitative variables, and not binary variables.

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Introduction to Ward's Method

- Based on the notion that clusters of multivariate observations should be approximately elliptical in shape, we assume that the data from each of the clusters have been realized in a multivariate distribution.
- Therefore, it would follow that they would fall into an elliptical shape when plotted in a p-dimensional scatter plot.

Notation

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Let X_{ijk} denote the value for variable k in observation j belonging to cluster i.

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Error Sum of Squares

$$SSE = \sum_i \sum_j \sum_k |X_{ijk} - \bar{x}_{i\cdot k}|^2$$

- We sum over all variables, and all of the units within each cluster.
- We compare individual observations for each variable against the cluster means for that variable.
- Note that when the Error Sum of Squares is small, it suggests that our data are close to their cluster means, implying that we have a cluster of like units.

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Total Sum of Squares

$$SST = \sum_i \sum_j \sum_k |X_{ijk} - \bar{x}_{..k}|^2$$

- The total sums of squares is defined in the same as always.
- Here we compare the individual observations for each variable against the grand mean for that variable.

R-Square

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$$R^2 = \frac{SST - SSE}{SST}$$

- This R^2 value is interpreted as the proportion of variation explained by a particular clustering of the observations.

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Strategy

- Using Ward's Method we start out with all sample units in n clusters of size 1 each.
- In the first step of the algorithm, $n-1$ clusters are formed, one of size two and the remaining of size 1.
- The error sum of squares and R^2 values are then computed.
- The pair of sample units that yield the smallest error sum of squares, or equivalently, the largest R^2 value will form the first cluster.

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Strategy

- Then, in the second step of the algorithm, $n-2$ clusters are formed from that $n-1$ clusters defined in step 2.
- These may include two clusters of size 2, or a single cluster of size 3 including the two items clustered in step 1.
- Again, the value of R^2 is maximized.
- Thus, at each step of the algorithm clusters or observations are combined in such a way as to minimize the results of error from the squares or alternatively maximize the R^2 value.
- The algorithm stops when all sample units are combined into a single large cluster of size n.

Ward's Method: Example

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We apply Ward's method to the Woodyard Hammock data.

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Introduction to K-Means Method

- This is a non-hierarchical approach.
- One of the advantages of this method is that we do not have to calculate the distance measures between all pairs of subjects.
- Therefore, this procedure seems much more efficient or practical when you have very large datasets.

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Introduction to K-Means Method

- Under this procedure you need to pre-specifiy how many clusters you want to consider.
- The clusters in this procedure do not form a tree.
- There are two approaches to carrying out the K-Means procedure.
- The approaches vary as to how the procedure begins the partitioning.
- The first approach is to do this randomly, to start out with a random partitioning of subjects into groups and go from there.
- The alternative is to start with an additional set of starting points. These would form the centers of our clusters.
- The random nature of the first approach will avoid bias.

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Overview of the Method

1. Partition the items into K initial clusters.
2. Scan through the list of n items, assigning each item to the cluster whose centroid (mean) is closest. Each time an item is reassigned we will recalculate the cluster mean or centroid for the cluster receiving that item and the cluster losing that item.
3. Repeat Step 2 over and over again until no more reassignments are made.

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Starting Point

Item	X_1	X_2
A	7	9
B	3	3
C	4	1
D	3	8

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Starting Point

Suppose that items are initially decide to partition the items into two clusters (A, B) and (C, D). Then calculating the cluster centroids, or the mean of all the variables within the cluster, we would obtain:

Cluster	$\bar{x}_1$	$\bar{x}_2$
(A,B)	$\frac{7+3}{2} = 5$	$\frac{9+3}{2} = 6$
(C,D)	$\frac{4+3}{2} = 3.5$	$\frac{1+8}{2} = 4.5$

For example, the mean of the first variable for cluster (A, B) is 5.

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Distance Calculations

Next, we calculate the distances between the item A and the centroids of clusters (A, B) and (C, D).

Cluster	Distance to A
(A,B)	$\sqrt{(7-5)^2+(9-6)^2} = \sqrt{13}$
(C,D)	$\sqrt{(7-3.5)^2+(9-4.5)^2} = \sqrt{32.5}$

- Here, we get a Euclidean distance between A and each of these cluster centroids.
- We see that item A is closer to cluster (A, B) than cluster (C, D).
- Therefore, we are going to leave item A in cluster (A, B) and no change is made at this point.

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Distance Calculations

Next, we will look at the distance between item B and the centroids of clusters (A, B) and (C, D).

Cluster	Distance to B
(A,B)	$\sqrt{(3-5)^2+(3-6)^2} = \sqrt{13}$
(C,D)	$\sqrt{(3-3.5)^2+(3-4.5)^2} = \sqrt{2.5}$

- Here, we see that item B is closer to cluster (C, D) than cluster (A, B).
- Therefore, item B will be reassigned, resulting in the new clusters (A) and (B, C, D).

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Centroids of the New Clusters

The centroids of the new clusters now changed are calculated as:

Cluster	$\bar{x}_1$	$\bar{x}_2$
(A)	7	9
(B,C,D)	$\frac{3+4+3}{2} = 3.3$	$\frac{3+1+8}{2} = 4$

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Distance Calculations

Next, we will calculate the distance between the items and each of the clusters (A) and (B, C, D).

Cluster	C	D	A	B
(A)	$\sqrt{73}$	$\sqrt{17}$	0	$\sqrt{52}$
(B,C,D)	$\sqrt{9.4}$	$\sqrt{16.1}$	$\sqrt{38.4}$	$\sqrt{1.1}$

- It turns out that since all four items are closer to their current cluster centroids, no further reassignments are required.

Sensitivity to Initial Clustering

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We must note however, that the results of the K-means procedure can be sensitive to the initial assignment of clusters.

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For example, suppose the items had initially been assigned to the clusters (A, C) and (B, D). Then the cluster centroids would be calculated as follows:

Cluster	$\bar{x}_1$	$\bar{x}_2$
(A,C)	$\frac{7+4}{2} = 5.5$	$\frac{9+1}{2} = 5$
(B,D)	$\frac{3+3}{2} = 3$	$\frac{3+8}{2} = 5.5$

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Distance Calculations

From here we can find that the distances between the items and the cluster centroids are:

Cluster	A	B	C	D
(A, C)	$\sqrt{18.25}$	$\sqrt{10.25}$	$\sqrt{18.25}$	$\sqrt{15.25}$
(B, D)	$\sqrt{28.25}$	$\sqrt{6.25}$	$\sqrt{21.25}$	$\sqrt{6.25}$

- Note that each item is closer to its cluster centroid than the opposite centroid.
- So, the initial cluster assignment is retained.

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Two Approaches

- Random assignment
- Leader algorithm

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Random Assignment

- The first approach is just to assign the clusters randomly.
- This does not seem like it would be a very efficient approach.
- The main reason to take this approach would be to avoid any bias in this process.

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Leader Algorithm

1. Select the first item from the list. This item will form the centroid of the initial cluster.
2. Search through the subsequent items until an item is found that is at least distance δ away from any previously defined cluster centroid. This item will form the centroid of the next cluster.
3. Step 2 is repeated until all K cluster centroids are obtained, or no further items can be assigned.
4. The initial clusters are obtained by assigning items to the nearest cluster centroids.

Example: K-Means Method

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Summary

In this unit we learned about ...

- The objectives of a cluster analysis;
- Methods for measuring distances or similarities between subjects;
- Linkage methods for measuring the distances between clusters;
- The difference between agglomerative and divisive clustering;
- How to interpret tree diagrams and select how many clusters are of interest;
- How to use individual ANOVAs and cluster means to describe cluster composition;
- The definition of Ward's method;
- The definition of the K-means method.

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What Is Discriminant Analysis?

Discriminant analysis is a classification problem, where two or more groups or clusters or populations are known a priori and one or more new observations are classified into one of the known populations based on the measured characteristics.

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Example 1: Swiss Bank Notes

We have two populations of bank notes, genuine, and counterfeit. Six measures are taken on each note:

- Length
- Right-hand width
- Left-hand width
- Top margin
- Bottom margin
- Diagonal across the printed area

Take a bank note of unknown origin and determine just from these six measurements whether or not it is real or counterfeit.

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Example 2: Pottery Data

Pottery shards are sampled from four sites: L) Llanedyrn, C) Caldicot, I) Ilse Thornes, and A) Ashley Rails and the concentrations of the following chemical constituents were measured at a laboratory.

- Al: Aluminum
- Fe: Iron
- Mg: Magnesium
- Ca: Calcium
- Na: Sodium

An archaeologist encounters a pottery specimen of unknown origin. To determine possible trade routes, the archaeologist may wish to classify its site of origin.

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Example 3: Insect Data

Data were collected on two species of insects in the genus Chaetocnema, (a) Ch. concinna and (b) Ch. heikertlingeri. Three variables were measured on each insect:

- width of the first joint of the tarsus (legs)
- width of the second joint of the tarsus
- width of the aedeagus

Our objective is to obtain a classification rule for identifying the insect species based on these three variables. An entomologist can identify these two closely related species, but the differences are so subtle that one has to have considerable experience to be able to tell the difference. If a classification rule may be developed, then this might be a more accurate way to help differentiate between these two different species.

Bayes' Rule

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Consider any two events A and B. To find $P(B|A)$, the probability that B occurs given that A has occurred, Bayes' Rule states the following:

$$P(B|A) = \frac{P(B \text{ and } A)}{P(A)}$$

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Bayes' Rule Applied to the Classification Problem

- We are interested in $P(\pi_i|\vec{x})$, the conditional probability that an observation came from population π_i given that the observed values of the multivariate vector of variables $\vec{x}$.
- We will classify an observation to the population for which the value of $P(\pi_i|\vec{x})$ is greatest.
- This is the most probable group given the observed values of $\vec{x}$.

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Bayes' Rule Applied to the Classification Problem

- Suppose that we have g populations (groups) and that the ith population is denoted as π_i.
- Let $p_i = P(\pi_i)$, the probability that a randomly selected observation is in population π_i.
- Let $f(\vec{x}|\pi_i)$ be the conditional probability density function of the multivariate set of variables x, given that observation came from population π_i.

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$$P(\text{ member of } \pi_i | \text{ we observed } \vec{x}) =$$

$$= \frac{P(\text{ member of } \pi_i \text{ and we observe } \vec{x})}{P(\text{ we observe } \vec{x})}$$

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$$P(\pi_i|\vec{x}) = \frac{p_i f(\vec{x}|\pi_i)}{\sum_{j=1}^{g} p_j f(\vec{x}|\pi_j)}$$

The **classification rule** is to assign observation $\vec{x}$ to the population for which the posterior probability is the greatest.

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The denominator is the same for all posterior probabilities (for the various populations) so it is equivalent to say that we will classify an observation to the population for which

$$p_i f(\vec{x}|\pi_i)$$

is greatest.

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Two Populations

With only two populations we can express a classification rule in terms of the ratio of the two posterior probabilities. Specifically we would classify to population 1 when

$$\frac{p_1 f(\vec{x}|\pi_1)}{p_2 f(\vec{x}|\pi_2)} > 1$$

i.e.

$$\frac{f(\vec{x}|\pi_1)}{f(\vec{x}|\pi_2)} > \frac{p_2}{p_1}$$

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Decision Rule

We are going to classify the sample unit or subject into the population π_i that maximizes the posterior probability $p(\pi_i)$, that is the population that maximizes

$$f(\vec{x}|\pi_i)p_i$$

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Decision Rule

- We are going to calculate the posterior probabilities for each of the populations.
- Then we are going to assign the subject or sample unit to that population that has the highest posterior probability.
- Ideally that posterior probability is going to be greater than a half, the closer to 100% the better.

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Decision Rule

Equivalently we are going to assign it to the population that maximizes this product:

$$\log f(\vec{x}|\pi_i)p_i$$

- The denominator that appears above does not depend on the population since it involves summing over all the populations.
- Equivalently all we really need to do is to assign it to the population that has the largest for this product, or equivalently again we can maximize the log of that product.
- A lot of times it is easier to write this log down.

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1. Collect ground truth or training data.
2. Estimation of prior probabilities.
3. Determine if variance-covariance matrices are homogeneous.
4. Estimate the parameters of the conditional probability density functions.
5. Compute discriminant functions.
6. Use cross validation to estimate misclassification probabilities.

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Step 1: Collect ground truth or training data

- *Ground truth* or *training data* are data with known group memberships.
- Here, we actually know to which population each subject belongs.
- For example, in the Swiss Bank Notes, we actually know which of these are genuine notes and which others are counterfeit examples.

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Step 2: Estimation of prior probabilities

The prior probability p_i represents the expected portion of the community that belongs to population π_i. There are three common choices:

- Equal priors
- Arbitrary priors
- Estimated priors

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Equal Priors

$$\hat{p}_i = \frac{1}{g}$$

This would be used if we believe that all of the population sizes are equal.

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Arbitrary priors

Arbitrary priors selected according to the investigators beliefs regarding the relative population sizes. Note

that we require:

$$\hat{p}_1 + \hat{p}_2 + \ldots + \hat{p}_g = 1$$

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Estimated priors

$$\hat{p}_i = \frac{n_i}{N}$$

where n_i is the number of observations from population π_i in the training data and $N = n_1 + n_2 + \ldots + n_g$.

Step 4: Determine if variance-covariance matrices are homogeneous

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- Use Bartlett's test to determine if variance-covariance matrices are homogeneous for the two or more populations involved.
- Result of this test will determine whether to use linear or quadratic discriminant analysis.

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Case 1: Linear Discriminant Analysis

Linear discriminant analysis is for homogeneous variance-covariance matrices:

$$\Sigma_1 = \Sigma_2 = \Sigma_g = \Sigma$$

In this case the variance-covariance matrix does not depend on the population from which the data are obtained.

Case 2: Quadratic Discriminant Analysis

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Quadratic discriminant analysis is used for heterogeneous variance-covariance matrices:

$$\Sigma_i \neq \Sigma_j \text{ for some } i \neq j$$

This allows the variance-covariance matrices to depend on which population we are looking at.

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Step 4: Estimate the Parameters of the Conditional Probability Density Functions

Estimate the parameters of the conditional probability density functions $f(\vec{X}|\pi_i)$. Here, we shall make the following standard assumptions:

1. The data from group i has common mean vector μ_i.
2. The data from group i has common variance-covariance matrix Σ.
3. Independence: The subjects are independently sampled.
4. Normality: The data are multivariate normally distributed.

Step 5: Compute Discriminant Functions

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This is the rule for classification of the new object into one of the known populations.

Step 6: Use cross validation to estimate misclassification probabilities.

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- As in all statistical procedures it is helpful to use diagnostic procedures to asses the efficacy of the discriminant analysis.
- We use cross-validation to assess the classification probability.
- Typically you are going to have some prior rule as to what is an acceptable misclassification rate.
- Those rules might involve things like, "what is the cost of misclassification?"
- There are really two alternative costs ...

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Linear Discriminant Analysis

We assume that in population n_i the probability density function of $\vec{x}$ is multivariate normal with mean vector μ_i and variance-covariance matrix Σ (same for all populations).

$$f(\vec{x}|\pi_i) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\left[-\frac{1}{2}(\vec{x}-\vec{\mu}_i)'\Sigma^{-1}(\vec{x}-\vec{\mu}_i)\right]$$

We classify to the population for which $p_i f(\vec{x}|\pi_i)$ is largest.

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Because a log transform is monotonic, this is equivalent to classifying an observation to the population for which

$$log(p_i f(\vec{x}|\pi_i))$$

is largest.

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Linear Score Function

- Linear discriminant analysis is used when the variance-covariance matrix does not depend on the population from which the data are obtained.
- In this case, our decision rule is based on the so-called *linear score function* which is a function of the population means for each of our g populations μ_i, as well as the pooled variance-covariance matrix.

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Linear Score Function

The linear score function is

$$s_i^L(\vec{X}) = -\frac{1}{2}\vec{\mu}_i'\Sigma^{-1}\vec{\mu}_i' + \vec{\mu}_i'\Sigma^{-1}\vec{x} + \log(p_i) =$$

$$= d_{i0} + \sum_{j=1}^{p} d_{ij}x_j + log(p_i)$$

where

$$d_{i0} = -\frac{1}{2}\vec{\mu}_i'\Sigma^{-1}\vec{\mu}_i'$$

and

$$d_{ij} = j^{th} \text{ element of } \vec{\mu}_i'\Sigma^{-1}$$

The far left-hand expression resembles a linear regression with intercept term d_{i0} and regression coefficients d_{ij}.

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Linear Discriminant Function

The linear discriminant function is:

$$d_i^L(\vec{X}) = -\frac{1}{2}\vec{\mu}_i'\Sigma^{-1}\vec{\mu}_i' + \vec{\mu}_i'\Sigma^{-1}\vec{x} =$$

$$= d_{i0} + \sum_{j=1}^{p} d_{ij}x_j$$

where

$$d_{i0} = -\frac{1}{2}\vec{\mu}_i'\Sigma^{-1}\vec{\mu}_i'$$

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Linear Discriminant Function

- Given a sample unit with measurements $x_1, x_2, \ldots, x_p$, we classify the sample unit into the population that has the largest linear score function.
- This is equivalent to classifying to the population for which the posterior probability of membership is largest.
- The linear score function is computed for each population, then we assign the unit to the population with the largest score.

Linear Discriminant Function

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However, the linear score function is a function of unknown parameters, $\vec{\mu}_i$ and Σ. So, these must be estimated from the data.

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Required Estimates

Discriminant analysis requires estimates of:

- Prior probabilities $p_i = P(\pi_i), i = 1, 2, \ldots, g$;
- The population means $\vec{\mu}_i$;
- The variance-covariance matrix Σ.

Typically, these parameters are estimated from training data, in which the population membership is known.

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The population means can be estimated by the sample mean vectors:

$$\vec{\mu}_i = E(\vec{X}|\pi_i) = \vec{\bar{x}}_i; \quad i = 1, 2, \ldots, g$$

Estimation of the Variance-Covariance Matrix

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The variance-covariance matrix can be estimated by using the pooled variance-covariance matri:x

$$\Sigma = var(\vec{X}|\pi_i); \quad i = 1, 2, \ldots, g$$

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Let S_i denote the sample variance-covariance matrix for population i. Then the variance-covariance matrix Σ can be estimated by substituting in the pooled variance-covariance matrix into the linear score function:

$$S_p = \frac{\sum_{i=1}^{g}(n_i - 1)S_i}{\sum_{i=1}^{g}(n_i - 1)}$$

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This can then be used to obtain the estimated linear score function:

$$\hat{s}_i^L(\vec{X}) = -\frac{1}{2}\vec{\bar{x}}_i' S_p^{-1}\vec{\bar{x}}_i + \vec{\bar{x}}_i' S_p^{-1}\vec{x} + \log(\hat{p}_i) =$$

$$= \hat{d}_{i0} + \sum_{j=1}^{p} \hat{d}_{ij} x_j + log(\hat{p}_i)$$

where

$$\hat{d}_{i0} = -\frac{1}{2}\vec{\bar{x}}_i' S_p^{-1}\vec{\bar{x}}_i'$$

and

$$\hat{d}_{ij} = j^{th} \text{ element of } \vec{\bar{x}}_i' S_p^{-1}$$

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Estimation of the Linear Score Function

- This is a function of the sample mean vectors, the pooled variance-covariance matrix and prior probabilities for g different populations.
- This is written in a form that looks like a linear regression formula with an intercept term plus a linear combination of response variables, plus the natural log of the prior probabilities.

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Decision Rule

Use as a decision rule:

Classify the sample unit into the population that has the largest estimated linear score function.

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Example: Insect Data

Data were collected on two species of insects in the genus Chaetocnema, (a) Ch. concinna and (b) Ch. heikertlingeri. Three variables were measured on each insect:

- Width of the first joint of the tarsus (legs)
- Width of the second joint of the tarsus
- Width of the aedeagus

Our objective is to obtain a classification rule for identifying the insect species based on these three variables.

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4. Estimate the parameters of the conditional probability density functions.
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Estimated Posterior Probabilities

The estimated posterior probability that the insect belongs to species (a) Ch. concinna given the observed measurements can be obtained by using this formula:

$$P(\pi_a|\vec{x}) = \frac{\exp\left[\hat{s}_a^L(\vec{x})\right]}{\exp\left[\hat{s}_a^L(\vec{x})\right] + \exp\left[\hat{s}_b^L(\vec{x})\right]} =$$

$$= \frac{\exp(202.359)}{\exp(202.359) + \exp(205.219)} = 0.05$$

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Estimated Posterior Probabilities

Similarly for species (b), the estimated posterior probability that the insect belongs to Ch. heikertlingeri is:

$$P(\pi_b|\vec{x}) = \frac{\exp\left[\hat{s}_b^L(\vec{x})\right]}{\exp\left[\hat{s}_a^L(\vec{x})\right] + \exp\left[\hat{s}_b^L(\vec{x})\right]} =$$

$$= \frac{\exp(205.219)}{\exp(202.359) + \exp(205.219)} = 0.95$$

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Estimating Misclassification Probabilities

- One procedure to measure how good the discriminant rule is, we classify the training data according to the developed discrimination rule.
- Since we know which unit comes from which population among the training data, this will give us some idea of the validity of the discrimination procedure.

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Estimating Misclassification Probabilities

- Method 1: The confusion table
- Method 2: Set aside method
- Method 3: Cross validation

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Method 1: The Confusion Table

- We will let:

$$p(i|j)$$

denote the probability that a unit from population π_j is classified into population π_i.

- The misclassification probabilities can be estimated by taking the number of insects from population j that are misclassified into population i divided by the total number of insects in the sample from population j as shown here:

$$\hat{p}(i|j) = \frac{n_{ji}}{n_{j.}}$$

This will give the misclassification probabilities.

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Example: Insect Data

From the statistics software output, we obtain the following confusion table:

Truth / Estimation	a	b	Total
a	10	0	10
b	0	10	10
Total	10	10	20

Here, no insect was misclassified. So, the misclassification probabilities are all estimated to be equal to zero.

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Method 2: Set Aside Method

1. Randomly partition the observations into two "halves".
2. Use one "half" to obtain the discriminant function.
3. Use the discriminant function from Step 2 to classify all members of the second "half" of the data, from which the proportion of misclassified observations can be computed.

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Method 2: Set Aside Method

- Advantage:
  This method yield unbiased estimates of the misclassification probabilities.
- Problem:
  Does not make optimum use of the data, and so, estimated misclassification probabilities are not as precise as possible.

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Method 3: Cross Validation

1. Delete one observation from the data.
2. Use the remaining observations to compute a discriminant function.
3. Use the discriminant function from Step 2 to classify the observation removed in Step 1. Steps 1-3 are repeated for all observations; compute the proportions of observations that are misclassified.

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Example: Insect Data

The confusion table for the cross validation is:

Truth / Estimation	a	b	Total
a	10	0	10
b	2	8	10
Total	12	8	20

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Example: Insect Data

Here, the estimated misclassification probabilities are:

$$\hat{p}(b|a) = \frac{0}{10} = 0$$

for insects belonging to species A, and

$$\hat{p}(a|b) = \frac{2}{10} = 0.2$$

for insects belonging to species B.

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Quadratic Discriminant Analysis

- Linear discriminant analysis is for homogeneous variance-covariance matrices.
- However, not in all cases data may come from such simplified situations.
- Quadratic discriminant analysis is used for heterogeneous variance-covariance matrices:

$$\Sigma_i \neq \Sigma_j \text{ for some } i \neq j$$

This allows the variance-covariance matrices to depend on which population we are looking at.

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Quadratic Score Function

$$s_i^Q(\vec{x}) = -\frac{1}{2}\log|\Sigma_i| - \frac{1}{2}(\vec{x}-\vec{\mu}_i)'\Sigma_i^{-1}(\vec{x}-\vec{\mu}_i) + \log(p_i)$$

This is a function of population mean vectors and the variance-covariance matrices for i_{th} group. Similarly, we will determine a separate quadratic score function for each of the groups.

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Quadratic Score Function

- This is of course a function of the unknown population mean vector for group i and the variance-covariance matrix for group i.
- These will have to be estimated from ground truth data. As before, we replace the unknown values of μ_i, Σ_i and p_i by their estimates to obtain the estimated quadratic score function as shown next.

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Decision Rule

Our decision rule remains the same as well. We will classify the sample unit or subject into the population that has the largest quadratic score function.

$$\hat{s}_i^Q(\vec{X}) = -\frac{1}{2}\log|S_i| - \frac{1}{2}(\vec{x} - \vec{\bar{x}})' S_i^{i-1}(\vec{x} - \vec{\bar{x}}) + \log(\hat{p}_i)$$

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Example: Swiss Bank Notes

We have two populations of bank notes, genuine, and counterfeit. Six measures are taken on each note:

- Length
- Right-hand width
- Left-hand width
- Top margin
- Bottom margin
- Diagonal across the printed area

Take a bank note of unknown origin and determine just from these six measurements whether or not it is real or counterfeit.

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Prior Probabilities

The prior probabilities can be expressed as:

$$\hat{p}_1 = 0.99$$

and

$$\hat{p}_2 = 0.01$$

Misclassification Probabilities: Cross Validation

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The confusion table for the cross validation is:

Truth / Estimation	a	b	Total
a	98	2	100
b	1	99	100
Total	99	101	200

Misclassification Probabilities: Cross Validation

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Here, we can see that 98 out of 100 counterfeit notes are expected to be correctly classified, while 99 out of 100 genuine notes are expected to be correctly classified. Thus, the estimated misclassification probabilities are estimated to be:

$$\hat{p}(real|fake) = 0.02$$

and

$$\hat{p}(fake|real) = 0.01$$

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Summary

In this unit we learned about ...

- The objectives of a discriminant analysis;
- How to determine the type of discriminant analysis to be carried out, linear or quadratic;
- How the linear discriminant function can be used to classify a subject into the appropriate population;
- Issues regarding the selection of prior probabilities that a randomly selected subject belongs to a particular population;
- The use of posterior probabilities to assess the uncertainty of the classification of a particular subject;
- The use of crossvalidation and confusion tables to assess the efficacy of discriminant analysis.

Canonical Correlation

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What Is Canonical Correlation Analysis?

Canonical correlation analysis is a method for exploring the relationships between two multivariate sets of variables (vectors), all measured on the same individual.

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Example 1

Consider as an example variables related to exercise and health.

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Example 1: Exercise Variables

- Climbing rate on a stair stepper
- Running speed
- Amount of weight lifted on bench press
- Number of push-ups per minute
- ...

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Example 1: Health Variables

- Blood pressure
- Cholesterol levels
- Glucose levels
- Body mass index
- ...

Example 1

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So two types of variables are measured and the relationships between the exercise variables and the health variables are to be studied.

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Example 2

Consider as an example variables related to environmental health and environmental toxins.

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Example 2: Environmental Health Variables

- Frequencies of sensitive species
- Species diversity
- Total biomass
- Productivity of the environment
- ...

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Example 2: Environmental Toxins Variables

- Concentrations of heavy metals
- Concentrations of pesticides
- Concentrations of dioxin
- ...

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The Use of Canonical Correlation Analysis

- To describe the relationship between the first set of variables and the second set of variables.
- Not thinking of one set as dependent and the other as independent.

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Motivation for Canonical Correlation Analysis

Alternative: Scatter plots

- It is possible to create pairwise scatter plots with variables in the first set (e.g., exercise variables), and variables in the second set (e.g., health variables).
- But if the dimension of the first set is p and that of the second set is q, there will be pq such scatter plots, it may be difficult, if not outright impossible, to look at all of these graphs together and be able to interpret the results.

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Alternative: Correlation analysis

- Similarly, you could compute all correlations between variables from the first set (e.g., exercise variables), and then compute all the correlations between the variables in the second set (e.g., health variables).
- But with pq a large number, problem of interpretation arises.

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Motivation for Canonical Correlation Analysis

- Canonical correlation analysis allows us to summarize the relationships into lesser number of statistics while preserving the main features of the relationships.
- In a way the motivation for canonical correlation is very similar to principal component analysis.
- It is also a dimension reduction technique.

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Starting Point

We have two sets of variables X and Y.
Let

$$\vec{X} = \begin{pmatrix} X_1 \\ X_2 \\ \dots \\ X_p \end{pmatrix}$$

and

$$\vec{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \dots \\ Y_q \end{pmatrix}$$

i.e. we have p variables in one set and q variables in the other set. We assume that $p \leq q$.

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Starting Point

- We look at linear combinations of the data (as in PCA).
- We define a set of linear combinations named U and V.
- U will correspond to the linear combinations from the first set of variables, X.
- V will correspond to the second set of variables, Y.
- Each member of U will be paired with a member of V.
- For example, U_1 (next slide) is a linear combination of the p X variables and V_1 is the corresponding linear combination of the q Y variables.

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Starting Point

Let

$$\begin{aligned} U_1 &= a_{11}X_1 + a_{12}X_2 + \cdots + a_{1p}X_p \\ U_2 &= a_{21}X_1 + a_{22}X_2 + \cdots + a_{2p}X_p \\ &\vdots \\ U_p &= a_{p1}X_1 + a_{p2}X_2 + \cdots + a_{pp}X_p \end{aligned}$$

and

$$\begin{aligned} V_1 &= b_{11}Y_1 + b_{12}Y_2 + \cdots + b_{1q}Y_q \\ V_2 &= b_{21}Y_1 + b_{22}Y_2 + \cdots + b_{2q}Y_q \\ &\vdots \\ V_p &= b_{p1}Y_1 + b_{p2}Y_2 + \cdots + b_{pq}Y_q \end{aligned}$$

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Canonical Variate Pair

We define

$$(U_i, V_i)$$

as the ith canonical variate pair.

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Canonical Variate Pair

- (U_1, V_1) is the first canonical variate pair.
- (U_2, V_2) is the second canonical variate pair.
- With $p \leq q$ there are p canonical covariate pairs.

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Objective

We are to find linear combinations that maximize the correlations between the members of each canonical variate pair.

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Variance of U_i

We can compute the variance of U_i variables using the following expression:

$$var(U_i) = \sum_{k=1}^{p} \sum_{l=1}^{p} a_{ik} a_{il} cov(X_k, X_l)$$

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Variance of U_i

- The coeffcients a_{i1} through a_{ip} that appear in the double sum are the same coefficients that appear in the definition of U_i.
- The covariances between the kth and I^{th} X-variables are multiplied by the corresponding coefficients a_{ik} and a_{il} for the variate U_i.

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Variance of V_i

Similar calculations can be made for the variance of V_j as shown below:

$$var(V_j) = \sum_{k=1}^{p} \sum_{l=1}^{q} b_{jk} b_{jl} cov(Y_k, Y_l)$$

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Covariance between U_i and V_j

$$cov(U_i, V_j) = \sum_{k=1}^{p} \sum_{l=1}^{q} a_{ik} b_{jl} cov(X_k, Y_l)$$

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Correlation between U_i and V_j

The correlation between U_i and V_j is calculated using the usual formula. We take the covariance between those two variables and divide it by the square root of the product of the variances:

$$corr(U_i, V_j) = \frac{cov(U_i, V_j)}{\sqrt{var(U_i)var(V_j)}}$$

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Canonical Correlation

The canonical correlation is a specific type of correlation. The canonical correlation for the ith canonical variate pair is simply the correlation between U_i and V_i :

$$\rho_i^* = \frac{cov(U_i, V_i)}{\sqrt{var(U_i)var(V_i)}}$$

Constraint:
This quantity is to be maximized. We want to find linear combinations of the X's and linear combinations of the Y's that maximize the above correlation.

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First Canonical Variate Pair: (U_1, V_1)

The coefficients $a_{11}, a_{12}, \ldots, a_{1p}$ and $b_{11}, b_{12}, \ldots, b_{1q}$ are to be selected so as to maximize the canonical correlation ρ_1^* of the first canonical variate pair. This is subject to the constraint that variances of the two canonical variates in that pair are equal to one:

$$var(U_1) = var(V_1) = 1$$

This is required so that unique values for the coefficients are obtained.

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Second Canonical Variate Pair: (U_2, V_2)

The coefficients $a_{21}, a_{22}, \ldots, a_{2p}$ and $b_{21}, b_{22}, \ldots, b_{2q}$ are to be selected so as to maximize the canonical correlation ρ_2^* of the second canonical variate pair. This is subject to the constraint that variances of the two canonical variates in that pair are equal to one:

$$var(U_2) = var(V_2) = 1$$

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Second Canonical Variate Pair: (U_2, V_2)

- Furthermore, we require the additional constraints that (U_1, U_2), and (V_1, V_2) have to be uncorrelated.
- In addition, the combinations (U_1, V_2) and (U_2, V_1) must be uncorrelated.

In summary, our constraints are:

$$cov(U_1, U_2) = cov(V_1, V_2) = 0$$

and

$$cov(U_1, V_2) = cov(U_2, V_1) = 0$$

Basically, we require that all of the remaining correlations equal zero.

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i^{th} Canonical Variate Pair: (U_i, V_i)

We want to find the coefficients $a_{i1}, a_{i2}, \ldots, a_{ip}$ and $b_{i1}, b_{i2}, \ldots, b_{iq}$ that maximizes the canonical correlation ρ_i^* subject to the similar constraints that

$$\begin{aligned}
var(U_i) &= var(V_i) = 1 \\
cov(U_1, U_i) &= cov(V_1, V_i) = 0 \\
cov(U_2, U_i) &= cov(V_2, V_i) = 0 \\
&\vdots \\
cov(U_{i-1}, U_i) &= cov(V_{i-1}, V_i) = 0 \\
cov(U_1, V_i) &= cov(U_i, V_1) = 0 \\
cov(U_2, V_i) &= cov(U_i, V_2) = 0 \\
&\vdots \\
cov(U_{i-1}, V_i) &= cov(U_i, V_{i-1}) = 0
\end{aligned}$$

Again, requiring all of the remaining correlations to be equal zero.

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The data to be analyzed comes from a firm that surveyed a random sample of $n = 50$ of its employees in an attempt to determine what factors influence sales performance. Two collections of variables were measured:
Sales performance:

- Sales Growth
- Sales Profitability
- New account sales

Test scores as a measure of intelligence :

- Creativity
- Mechanical reasoning
- Abstract reasoning
- Mathematics

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There are 3 variables in the first group relating to sale performance and 4 variables in the second group relating to the test scores.

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Strategy:

1. Test for relationship between canonical variate pairs
2. Obtain estimates of canonical correlation
3. Obtain the canonical coefficients
4. Interpret each component
5. Reinforcing the results

Test for Relationship Between Canonical Variate Pairs

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- The very first thing to determine is if there is any relationship between the two sets of variables at all.
- Perhaps the two sets of variables are completely unrelated to one another and independent!

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Test for Relationship Between Canonical Variate Pairs

- To test for independence between the sales performance and the test score variables first consider a multivariate multiple regression model where we are predicting, in this case, sales performance variables from the test score variables.
- In this general case, we are going to have p multiple regressions, each multiple regression predicting one of the variables in the first group (X variables) from the q variables in the second group (Y variables).

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Test for Relationship Between Canonical Variate Pairs

$$
\begin{array}{rcccccccc}
X_1 & = & \beta_{10} + \beta_{11} Y_1 & + & \beta_{12} Y_2 & + & \cdots & + & \beta_{1p} Y_q + \varepsilon_1 \\
X_2 & = & \beta_{20} + \beta_{21} Y_1 & + & \beta_{22} Y_2 & + & \cdots & + & \beta_{2p} Y_q + \varepsilon_2 \\
 & \vdots & & & & & & & \\
X_p & = & \beta_{p0} + \beta_{p1} Y_1 & + & \beta_{p2} Y_2 & + & \cdots & + & \beta_{pq} Y_q + \varepsilon_p
\end{array}
$$

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Test for Relationship Between Canonical Variate Pairs

- In our example, we have multiple regressions predicting the $p = 3$ sales variables from the $q = 4$ test score variables.
- We wish to test the null hypothesis that these regression coefficients (except for the intercepts) are all equal to zero.
- This would be equivalent to the null hypothesis that the first set of variables is independent from the second set of variables.

$$H_0 : \beta_{ij} = 0 \quad i = 1, 2, \ldots, p, \quad j = 1, 23, \ldots, q$$

This is carried out using Wilk's lambda.

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The previous null hypothesis is equivalent to testing the null hypothesis that all p canonical variate pairs are uncorrelated:

$$H_0 : \rho_1^* = \rho_2^* = \cdots = \rho_p^* = 0$$

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Result in the Example

Since Wilk's lambda is significant, and since the canonical correlations are ordered from largest to smallest, we can conclude that at least

$$\rho_1^* \neq 0$$

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Test for Relationship Between Canonical Variate Pairs

- We may also wish to test the null hypothesis that maybe the second or the third canonical variate pairs are correlated.
- We can do this in successive tests.
- Next, test whether the second and third canonical variate pairs are correlated.

$$H_0 : \rho_2^* = \rho_3^* = 0$$

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Result in the Example

Since Wilk's lambda is significant, and since the canonical correlations are ordered from largest to smallest, we can conclude that the second canonical variate pair is correlated:

$$\rho_2^* \neq 0$$

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- Finally, we can test the significance of the third canonical variate pair.

$$H_0 : \rho_3^* = 0$$

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Result in the Example

Since Wilk's lambda is significant, we can conclude that the third canonical variate pair is correlated:

$$\rho_3^* \neq 0$$

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Summary

In this unit we learned about ...

- The objectives of a canonical correlation analysis;
- How to test for independence between two sets of variables;
- How to determine the number of significant canonical variate pairs;
- How to compute the canonical variates from the data;
- How to interpret each member of a canonical variate pair using its correlations with the member variables;
- How to use the results of canonical correlation analysis to describe the relationships between two sets of variables.